

Why the geometry of state space is non-Archimedean

p-Adic Topology | Version 1.5

ZP Companion | April 2026

This companion explains the ideas in plain language with diagrams and real-world examples. It is not the formal ontology — every claim here restates a result already proven in the corresponding technical document. Consult that document for the authoritative mathematics.

Why p-Adic Geometry?

The standard choice for modeling distance — the real number line — is the wrong tool for binary existence. On the real line, between any two distinct points there is always a third: you can be "halfway between" existing and not existing. AX-B1 rules this out. A state either exists or it does not; there is no intermediate.

The 2-adic metric inverts this. In $\mathbb{Q}_2$, the two ontological states — null (0) and first atomic state (1) — are at maximum distance from each other: $d(0, 1) = 1$. More importantly, there is no point between them in the 2-adic topology. You cannot travel from 0 to 1 through a continuous path of intermediate values — the geometry has no room for partial existence. The binary distinction is enforced by the geometry itself, not imposed on top of it. This is why p-adic topology — specifically $\mathbb{Q}_2$ — is the natural mathematical language for a framework built on AX-B1. Note: this framework makes no claims about physical cosmology. The emergence described is formal — it concerns the logical structure of state in a mathematical ontology, not a theory of how the physical universe began.

The key property

Mathematical object: 2-adic valuation $v_2(x)$.

Ontological function: depth of binary structure — how many times 2 divides x .

The crucial point: $d(0, 1) = 1$ in $\mathbb{Q}_2$, with no path through an intermediate value. You cannot be half-way between existing and not existing.

What Is ZP-B Doing?

ZP-B puts the Zero Paradox on a specific geometric foundation: the 2-adic number field $\mathbb{Q}_2$. This is not standard Euclidean geometry. In $\mathbb{Q}_2$, distance works in a fundamentally different way — one that makes the singularity at zero structurally unavoidable.

The starting point is a single axiom: AX-B1 (Binary Existence) — a state either exists or it does not. From this, together with a minimality principle (MP-1), the document derives that the field must be $\mathbb{Q}_2$. The choice of geometry is proven as Theorem T0, not assumed.

Real-world example — Light switch vs. dimmer

AX-B1 says existence is like a light switch: on or off, no dimmer. This is the foundational claim. From "binary only," the 2-adic geometry follows mathematically.

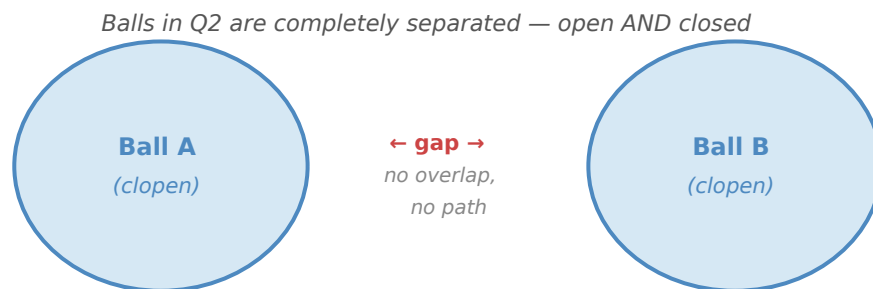
What Does "Non-Archimedean" Mean?

The title uses the term non-Archimedean, which simply means: the usual rule about distance does not apply. In ordinary geometry (Archimedean), if you take enough small steps you can travel any large distance — pile up enough pennies and you reach any amount. A non-Archimedean geometry breaks this: adding more small steps does not necessarily get you further. Distance is measured by a completely different rule — one based on divisibility by 2, not on cumulative size.

The practical consequence: in $\mathbb{Q}_2$, two points can be "far apart" even if their numerical difference looks small, and "close together" even if their numerical difference looks large. Closeness is about shared binary structure, not about proximity on the number line.

The Ultrametric: A Different Kind of Distance

In ordinary geometry the triangle inequality says: A-to-C $\leq$ A-to-B plus B-to-C. In $\mathbb{Q}_2$ a stronger law holds: A-to-C $\leq$ maximum of the two. This is the ultrametric. Every ball is simultaneously open and closed (clopen). There are no gradual boundaries — regions are entirely inside or entirely outside.



Two disjoint balls in $\mathbb{Q}_2$. They cannot overlap or gradually merge — completely separated. Every ball is both open and closed.

Real-world example — File folders on a computer

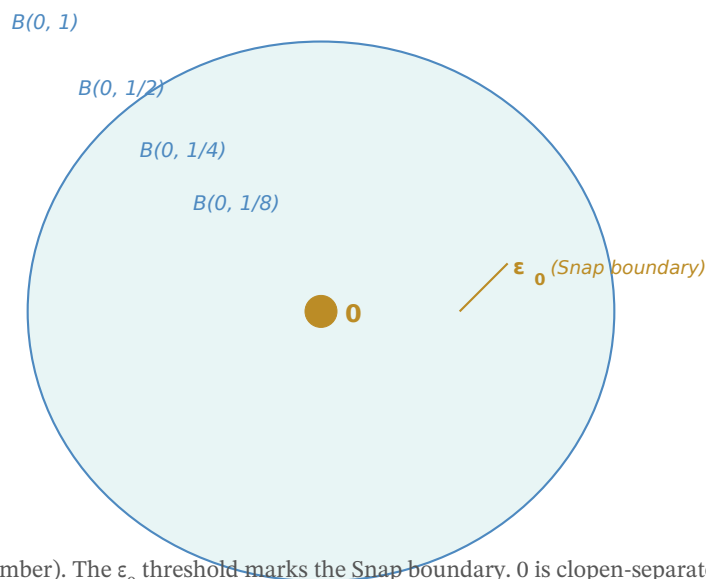
A file is either in a folder or it isn't — there's no "halfway in." Two files in completely different folder trees are maximally separated regardless of how similar their names are. This is ultrametric thinking: location in a hierarchy determines distance, not surface similarity.

The Ball Hierarchy Around Zero

The element $0 \in \mathbb{Q}_2$ is surrounded by nested balls $B(0,1)$, $B(0,1/2)$, $B(0,1/4)$, ... converging on 0. But 0 itself remains isolated. To understand why, we need the concept of a 2-adic valuation.

The 2-adic valuation of a number counts how many times 2 divides it. For example: $12 = 4 \times 3 = 2^2 \times 3$, so its valuation is 2. The number 7 is not divisible by 2 at all, so its valuation is 0. In $\mathbb{Q}_2$, distance between two numbers is determined by their valuation: the more times 2 divides their difference, the closer they are. This is the opposite of ordinary intuition — high divisibility means proximity.

The element 0 has a special status: it is divisible by 2 an unlimited number of times ($0 = 2 \times 0 = 4 \times 0 = \dots$ always). Its valuation is therefore infinite. This means 0 is at infinite "depth" in the hierarchy — no finite sequence of steps from the outside ever reaches it.



Balls in $\mathbb{Q}_2$ converging on 0 (amber). The ϵ_0 threshold marks the Snap boundary. 0 is clopen-separated — no continuous path from outside ever reaches it.

Key Result: Total Disconnectedness (T5)

The only connected subsets of $\mathbb{Q}_2$ are single points. There are no curves, no paths, no continuous structures bridging regions. This is proven from the clopen ball structure — a theorem, not an assumption.

Key Result: The Snap is Topologically Irreversible (C3)

There is no continuous path from any non-zero point back to 0. This follows directly from total disconnectedness (T5). Irreversibility is a corollary of the geometry, not a postulate.

Real-world example — Crossing a river with no bridge

Imagine a world where rivers have no bridges and no fords. Once you're on one bank, there is literally no continuous path to the other. That's total disconnectedness. 0 is on one side; every non-zero state is on the other. The Snap is the only crossing, and it only goes one way.

Remember: The 2-adic structure — ultrametric, clopen balls, clopen separation of 0 — is universal across all instantiations. The numerical value of ϵ_0 (the Snap threshold) is universe-contingent: it depends on physical constants. The Zero Paradox is a structural ontology, not a physical theory of our particular universe.